Amplitude-Encoding Kernels for Zero-Trust Network Anomaly Detection

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***Abstract*—Amplitude encoding is the most common way to load data into quantum kernels proposed for intrusion detection. Its fidelity kernel equals the squared cosine similarity of the inputs, so it can be computed classically at any scale. We compare it with classical kernels, tree ensembles and neural networks. Every model is trained on the same data at every training size. We use a synthetic zero-trust flow simulator, UNSW-NB15 and CIC-IDS2017. In distribution, the amplitude-encoding SVM (AE-SVM) ties an RBF-kernel SVM and trails tree ensembles. On attack types held out from training, AE-SVM fails on synthetic flows, with mean AUC 0.526 against 0.839 for RBF-SVM. Controls show the cause is the kernel's invariance to input norm. Appending one constant coordinate before encoding restores the AUC to 0.793 and also raises in-distribution accuracy. On the public benchmarks the failure does not appear. Mean held-out AUC is 0.945 against 0.932 for RBF-SVM on UNSW-NB15 and 0.715 against 0.658 on CIC-IDS2017. Held-out-attack tests on real traffic are therefore needed before judging such kernels.**

***Index Terms*—Quantum kernels, amplitude encoding, intrusion detection, zero trust, zero-day attacks**

I. Introduction

Zero-trust architectures check every flow before granting access [1], [2]. This makes flow-level anomaly detection part of each access decision.

Quantum kernel methods map inputs to quantum states and use state overlaps as a kernel [3], [4]. They have been applied to intrusion detection [5] and botnet detection [6]. Such evaluations often use a single operating point. The quantum model is often trained on fewer samples than the classical baselines, because circuit simulation is slow.

We test amplitude encoding under a matched protocol. Every model sees the same training data at every size. We also hold out whole attack types, since a deployed detector will meet attacks it was not trained on [7].

Our contributions are as follows.

1) A matched comparison on one simulator and two public benchmarks (Tables II and III).
2) A zero-day failure of AE-SVM on synthetic flows, traced to norm invariance by controlled interventions (Tables IV and V).
3) A one-coordinate norm anchor that repairs the failure without tuning on held-out attacks.
4) Evidence that the failure does not occur on UNSW-NB15 or CIC-IDS2017.

TABLE I
Datasets. Public data uses 20,000 training and 50,000 test flows per seed.

Dataset	Flows	Features	Attack types	Seeds
Synthetic flows	8,000	64	5	10
UNSW-NB15	257,673	68	9	5
CIC-IDS2017	2,520,730	69	6	5

II. The Amplitude-Encoding Kernel

Amplitude encoding pads $\mathbf{x} \in \mathbb{R}^d$ with zeros to 2^q entries, with $q = \lceil \log_2 d \rceil$. It then normalizes the vector into the state $|\psi(\mathbf{x})\rangle$. The fidelity kernel is

$$K_A(\mathbf{x}, \mathbf{y}) = |\langle \psi(\mathbf{x}) | \psi(\mathbf{y}) \rangle|^2 = \frac{(\mathbf{x}^\mathsf{T}\mathbf{y})^2}{\|\mathbf{x}\|^2 \, \|\mathbf{y}\|^2}. \tag{1}$$

It costs $\mathcal{O}(d)$ per pair on a classical computer. No quantum hardware is needed.

Eq. (1) gives $K_A(c\,\mathbf{x}, \mathbf{y}) = K_A(\mathbf{x}, \mathbf{y})$ for any $c \neq 0$. So a classifier built on K_A scores $\mathbf{x}$, $2\mathbf{x}$ and $-\mathbf{x}$ the same. On standardized features the origin is the training mean. The kernel therefore cannot use how far a flow lies from typical traffic.

Norm anchor. We append a constant before encoding,

$$\phi_c(\mathbf{x}) = \frac{(\mathbf{x},\, c)}{\sqrt{\|\mathbf{x}\|^2 + c^2}}, \qquad c = \mathrm{median}_{\mathrm{train}} \, \|\mathbf{x}\|. \tag{2}$$

The last amplitude now falls as $\|\mathbf{x}\|$ grows. The sign of $\mathbf{x}$ is no longer ambiguous. The median rule fixes c from training data alone.

Sensitivity. The Jacobian of $\mathbf{x} \mapsto \mathbf{x}/\|\mathbf{x}\|$ has spectral norm exactly $1/\|\mathbf{x}\|$. Normalization shrinks perturbations of large inputs and magnifies those near the origin. It gives no uniform bound on sensitivity.

Classifier. We train SVMs on K_A and on classical kernels with the same settings ($C = 1$, balanced class weights). Above 1,500 training samples we use the Nyström approximation [8]. Platt scaling gives probabilities [9]. Randomized smoothing [10] gives certified ℓ_2 radii.

III. Experimental Setup

Table I lists the data. The synthetic flows follow NetFlow-like marginals coupled by a Gaussian copula. Each of five

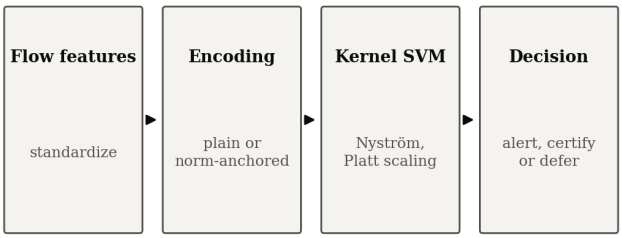


Fig. 1. Pipeline. Flow features are encoded, classified by a Nyström kernel SVM, certified by randomized smoothing and deferred to an analyst when uncertain.

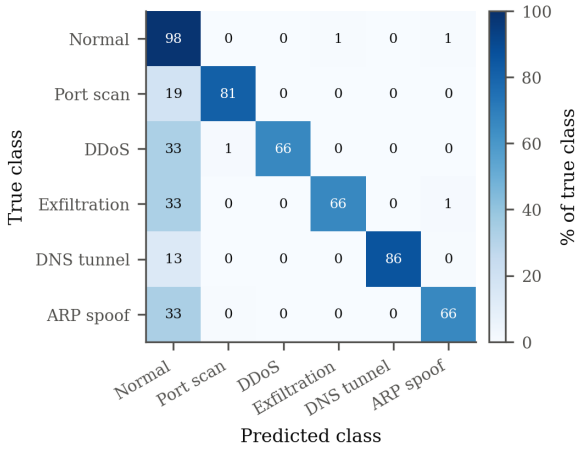


Fig. 2. Row-normalized confusion matrix of a one-vs-rest AE-SVM on synthetic flows. Most errors label attacks as normal.

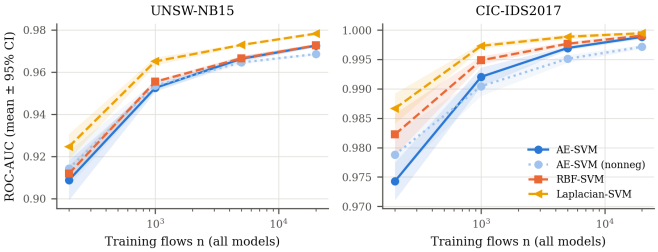


Fig. 3. Public data. ROC-AUC with every kernel trained on the same subsample at each n.

attacks shifts 6 to 7 latent dimensions. UNSW-NB15 [11] uses its official train and test split. CIC-IDS2017 [12] is cleaned of missing values, infinite values and 308,381 duplicates. Classes with fewer than 100 flows are dropped. The rest are grouped into six families with a fixed 70/30 split. Public features pass through $\operatorname{sign}(x)\log(1+|x|)$. All features are standardized on the training data.

Baselines are RBF, Laplacian and cubic polynomial SVMs, random forest (RF) [13], XGBoost [14] and a small MLP. On public data we add a 3×256 deep network (DNN) and an autoencoder trained on benign flows only, in the style of Kitsune [15]. We report means with 95% confidence intervals over seeds.

IV. Results

A. In-distribution detection

Table II shows that AE-SVM and RBF-SVM are tied. On synthetic flows their accuracies are 0.947 and 0.948 (paired t-test, $p = 0.54$). On UNSW-NB15 the gap is 0.001 ($p = 0.24$). On CIC-IDS2017 RBF-SVM is ahead by 0.001 ($p = 0.001$). Tree ensembles and neural networks lead on every dataset. The anchored encoding is the best kernel on synthetic flows. It matches plain AE-SVM on public data.

Fig. 2 shows where AE-SVM errs on synthetic flows. Between 13% and 33% of each attack class is labelled normal. Confusion between attack types is near zero. Per-class ROC-AUC still ranges from 0.930 to 0.993. A better threshold would recover part of the missed recall.

B. Training size and dimensionality

AE-SVM is never reliably ahead at any n or d (Table III). On synthetic flows it needs more data than RBF-SVM to match it. The gap also widens again at $d = 256$. Fig. 3 shows the same pattern on public data. The Laplacian kernel is the strongest kernel at every n.

C. Unseen attack types

Table IV holds the main result. On synthetic flows plain AE-SVM fails on every held-out type (Fig. 4). Its AUC lies between 0.45 and 0.61, while RBF-SVM stays between 0.81 and 0.89.

The two controls isolate the cause. Scaling inputs to $[0, 1]$ removes the sign ambiguity but keeps norm invariance. It helps little. The norm anchor keeps the norm. It closes most of the gap to RBF-SVM.

The public benchmarks behave differently (Fig. 5). On average AE-SVM matches or beats RBF-SVM on held-out families. On CIC-IDS2017 most supervised models struggle on Bot and PortScan. The benign-only autoencoder, which never sees an attack, is the strongest detector there.

Table V varies c. Small values leave the norm invisible. Very large values squeeze all inputs toward one point. The median rule lands near the best value without looking at held-out attacks.

Kernel-target alignment is lower for K_A than for RBF on every dataset (Table VI). It predicts the in-distribution ranking. It does not predict the held-out results, which depend on the kernel's invariances.

D. Certification, uncertainty and cost

Certified radii are close for both kernels (Table VII). AE-SVM abstains slightly more often. This fits the sensitivity result of Section II. Normalization gives the kernel no extra protection against noise.

TABLE II
IN-DISTRIBUTION DETECTION (MEAN ± 95% CI). BOLD MARKS THE BEST MEAN PER COLUMN.

	Synthetic flows		UNSW-NB15		CIC-IDS2017	
Model	Accuracy	AUC	F1	AUC	F1	AUC
AE-SVM	.947±.002	.966±.003	.887±.003	.973±.001	.979±.001	.999±.000
AE-SVM (anchored)	.960±.003	.983±.002	.884±.003	.972±.001	.977±.001	.998±.000
AE-SVM (nonneg)	–	–	.877±.003	.969±.001	.970±.002	.997±.000
RBF-SVM	.948±.003	.976±.003	.887±.003	.973±.001	.983±.001	.999±.000
Laplacian-SVM	.946±.003	.976±.002	.899±.004	.978±.001	.987±.001	**.999±.000**
Poly-SVM	.959±.003	.979±.003	.889±.003	.975±.001	.989±.001	.999±.000
RF	.943±.003	.976±.002	.889±.003	.980±.001	.990±.002	**.999±.000**
XGBoost	.962±.003	**.985±.002**	.892±.003	**.981±.001**	**.994±.001**	**1.000±.000**
MLP	**.965±.001**	.984±.002	.889±.004	.975±.001	.991±.001	**.999±.000**
DNN	–	–	**.909±.006**	.977±.002	.989±.002	**.999±.000**
Autoencoder	–	–	.394±.187	.887±.009	.546±.058	.948±.012

TABLE III
SYNTHETIC FLOWS. ACCURACY OF AE-SVM MINUS RBF-SVM (PERCENTAGE POINTS) AT MATCHED TRAINING SIZE n ($d = 64$) AND MATCHED DIMENSIONALITY d ($n = 2{,}800$).

n	50	100	200	500	1000	2000	5000
Gap	−3.6	−2.3	−1.5	−0.6	−0.2	−0.6	+0.2
d	16	32	64	128	256		
Gap	−3.1	−1.7	+0.5	−0.7	−4.5		

TABLE IV
MEAN ROC-AUC ON HELD-OUT ATTACK TYPES (LEAVE ONE TYPE OR FAMILY OUT). SYNTHETIC USES 5 SEEDS AND PUBLIC DATA 3 SEEDS. A DASH MEANS NOT RUN.

Model	Synthetic	UNSW-NB15	CIC-IDS2017
AE-SVM	0.526	0.945	0.715
AE-SVM (nonneg)	0.593	0.935	0.663
AE-SVM (anchored)	0.793	0.940	0.733
RBF-SVM	**0.839**	0.932	0.658
RF	–	0.949	0.820
XGBoost	–	0.960	0.852
DNN	–	**0.963**	0.694
Autoencoder	–	0.860	**0.874**

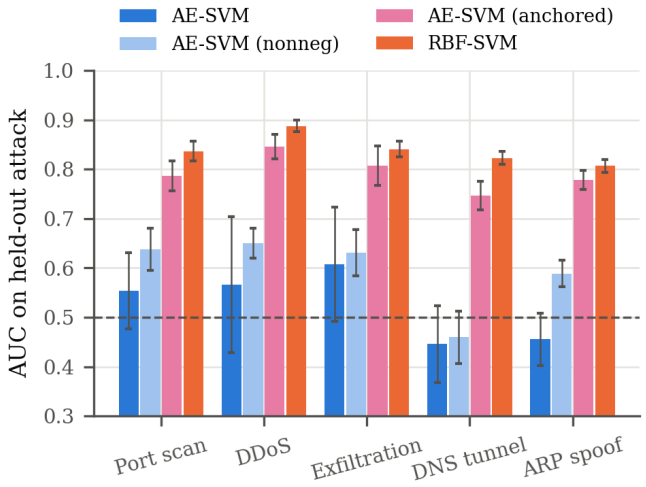


Fig. 4. Synthetic flows. AUC per held-out attack type (mean ± 95% CI, 5 seeds).

TABLE V
SYNTHETIC FLOWS. MEAN HELD-OUT AUC AGAINST THE ANCHOR CONSTANT c (5 SEEDS). $c = 0$ IS PLAIN ENCODING. THE MEDIAN RULE GIVES $c \approx 6.4$.

c	0	1	2	4	8	16	32	median
AUC	0.526	0.549	0.691	0.809	0.771	0.650	0.521	0.793

Deferring uncertain flows trades coverage for accuracy (Fig. 6). With $u < 0.31$, AE-SVM keeps 75.5% of flows at 97.4% accuracy. With $u < 0.10$, it keeps 26.8% at 99.9%.

All models run far below a per-flow budget (Table VIII). With few landmarks AE-SVM is slightly faster than RBF-SVM. With 1000 landmarks every kernel SVM is slower than XGBoost, the MLP and the autoencoder.

V. DISCUSSION AND LIMITATIONS

AE-SVM sees direction and ignores magnitude. This helps when magnitude is noise. It hurts when unseen attacks differ from normal traffic mainly in size, as in our simulator. On real traffic, attack families also differ in which features are active. There the weakness did not appear. A simulator alone would have led us to the wrong general conclusion.

Our companion study of QKD telemetry finds the same mechanism [17]. There, AE-SVM generalizes only while one feature carries magnitude.

The study has limits. The simulator is not packet-level traffic. Both public datasets have known labelling artifacts. We subsample them to 20,000 training flows, which favours kernel methods. Kernel settings were not tuned per dataset. Certified radii cover ℓ_2 noise only, not adaptive evasion.

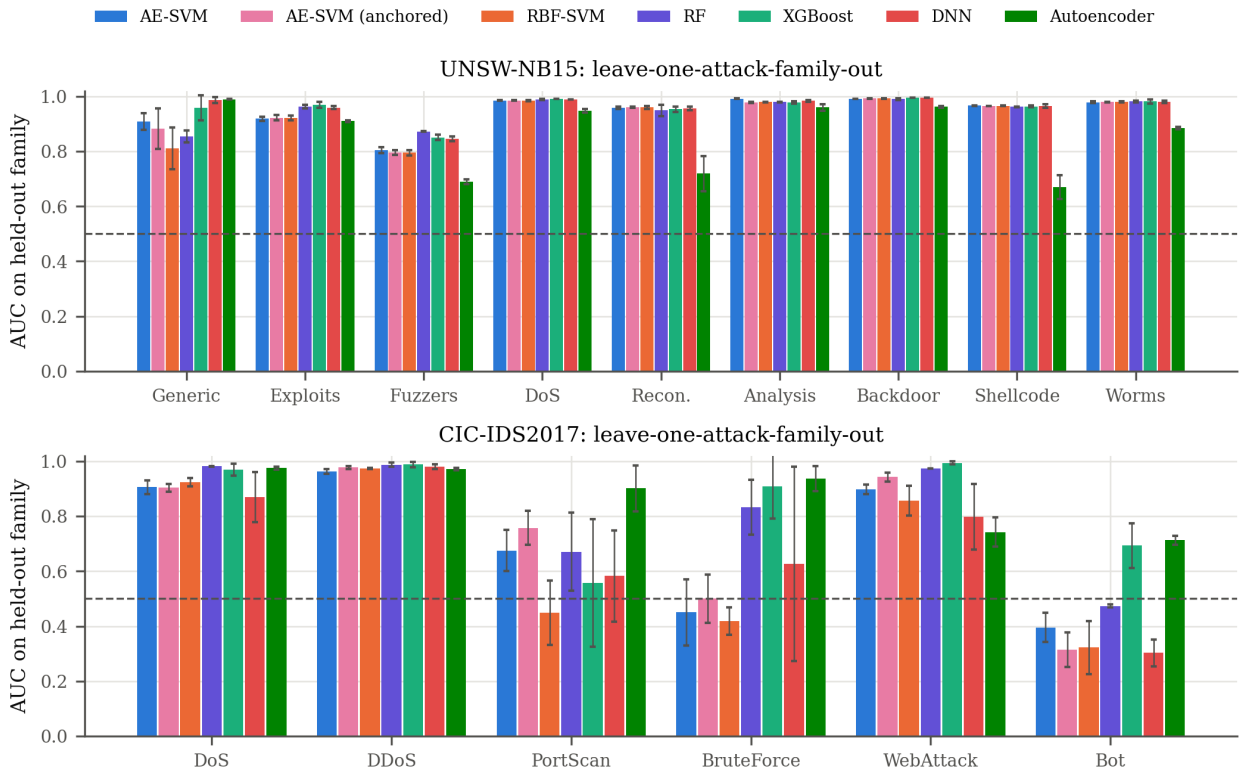


Fig. 5. Public data. AUC per held-out attack family (mean ± 95% CI, 3 seeds).

TABLE VI
KERNEL-TARGET ALIGNMENT [16], AVERAGED OVER ATTACK TYPES.

Kernel	Synthetic	UNSW-NB15	CIC-IDS2017
AE	0.13	0.38	0.62
AE (nonneg)	–	0.54	0.76
RBF	0.21	0.52	0.67
Laplacian	0.19	0.57	0.82

TABLE VII
SYNTHETIC FLOWS. RANDOMIZED-SMOOTHING CERTIFICATION ($\sigma = 0.25$, 300 NOISE DRAWS, $\alpha = 0.05$, 3 SEEDS).

Kernel	Mean radius	Median radius	Abstained
AE-SVM	0.508	0.582	1.8%
RBF-SVM	0.518	0.582	0.0%

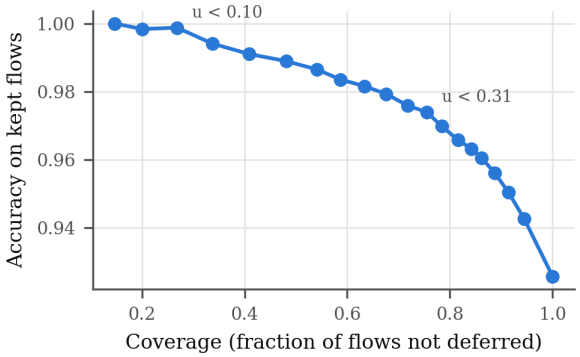


Fig. 6. Synthetic flows. Coverage and accuracy of Platt-scaled AE-SVM as the deferral threshold on $u = \sqrt{p(1-p)}$ varies.

VI. CONCLUSION

The amplitude-encoding kernel gives no in-distribution gain over an RBF kernel on flow data. It needs more data at small n. On synthetic flows it fails on unseen attacks because it discards the input norm. A one-coordinate norm anchor repairs this. On UNSW-NB15 and CIC-IDS2017 the failure does not occur. Quantum and quantum-style detectors should be tested on held-out attack families from real traffic, with their invariances checked first.

CODE AND DATA AVAILABILITY

Code, data preparation scripts and all result files are at https://github.com/arnavk23/quasar-security.

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TABLE VIII

BATCH INFERENCE COST PER FLOW (MS) ON ONE CPU CORE. SYNTHETIC KERNELS USE 50 NYSTRÖM LANDMARKS AND PUBLIC KERNELS 1000. TIMINGS COME FROM A SHARED MACHINE AND SHOW ORDERING ONLY.

Model	Synthetic	UNSW-NB15	CIC-IDS2017
AE-SVM	0.0048	0.1182	0.3391
RBF-SVM	0.0056	0.1339	0.3971
RF	0.0623	0.0603	0.0726
XGBoost	–	0.0273	0.0118
MLP	0.0016	0.0026	0.0011
DNN	–	0.0183	0.0169
Autoencoder	–	0.0046	0.0008

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